FGCU AI DAY 2026 Florida Gulf Coast University

Robotic Chess Engine

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A robot arm that can play chess.

HOW IT WORKS

1. You move a piece.
2. An overhead webcam classifies each square as green, blue or empty and matches the change against the legal moves.
3. Stockfish on the Raspberry Pi picks a reply (Elo 800 to 2000).
4. A Dobot Magician arm with an air-powered claw lifts the piece, travels above the other pieces and sets it down. Captured pieces go to trays.
5. An ESP32 touchscreen runs setup, calibration and the game.

HARDWARE

Raspberry Pi 5: where everything runs. The whole game is hosted on this Pi: the webcam vision, the chess rules, Stockfish, and every command sent to the arm and the touchscreen.



Raspberry Pi 5

- **Dobot Magician arm** with an air-powered claw
- **ESP32-S3 touchscreen** in a 3D-printed enclosure
- **Overhead webcam** watches all 64 squares
- **3D-printed pieces** in FGCU blue and green

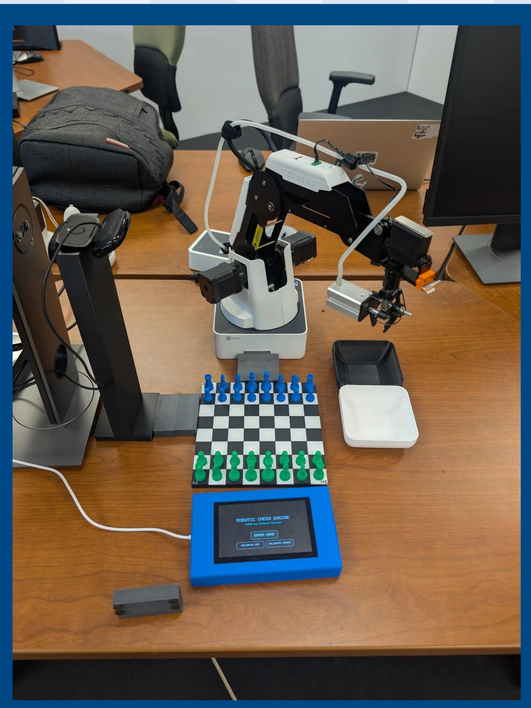
CALIBRATION

Arm. Jog the claw to A1, H1, A8 and H8 and tap CAPTURE. The Pi fits one perfect square grid through the four points. The same screen sets the capture trays, a high park spot and the table height.

Camera. The Pi sends a snapshot to the touchscreen. You tap the four board corners. The image is warped to a top-down board and cut into 64 cells.



The touchscreen



The whole table: webcam on the stand, arm, board, capture trays and touchscreen

HOW THE ARM MOVES A PIECE

1. **Park.** Between moves the arm waits high and to the side, so the webcam sees the whole board.
2. **Up.** The claw opens and the arm rises into a travel lane 50 mm above the board, high enough to clear a king.
3. **Across.** It goes straight to the square at that height, never dragging over other pieces.
4. **Down and grip.** Every kind of piece has its own grip height, so the claw closes on a pawn and on a king in the right place. A short pause first lets the arm stop shaking.
5. **Up, across, down, release.** The same trip to the new square, then the claw opens and the arm climbs back into the lane.
6. **Captures first.** The taken piece goes to a tray, spread over six spots so the pile never topples.



The Dobot Magician and its air-powered claw. The box in front is the air pump.

PITFALLS WE HIT

- **Drift.** The arm's coordinates are inconsistent and slowly go wrong as a game goes on: after 30 to 40 moves it starts to miss. Steps are lost when it is stopped short. Still open.
- **A small reach, limited in code.** By hand the arm could be pushed to spots it refused to drive to in code, worst of all the row closest to its base. Its working range is very small, and up high it reaches even less, so that row is entered sideways along its file.
- **The claw.** The air-pump claw opens an uncontrollable amount, so a zip tie now limits how far it opens.
- **Black on black.** The camera could not tell black and white pieces from the black and white squares. The pieces were printed blue and green.
- **The first grip.** It did nothing until the pump had cycled.
- **Captures.** If a removal missed, the arm stacked its piece on the other. Now the camera checks the square first.

WHERE THE AI HELPED

- **Writing the code.** The Pi script, the touchscreen firmware and the vision pipeline.
- **Fast testing.** Dry-run simulators (a fake arm and a fake camera) let us play whole games with no hardware on the desk.
- **Debugging from evidence.** One photo of the terminal, "the arm stopped at 181 mm from its base when asked for 148", gave away the reach limit.
- **Designing for recovery.** Every bad moment in a game has a way back.

WHERE IT DID NOT

- **A niche arm.** The AI did not know the Dobot protocol or its library. The team had to find the documentation and feed it in.
- **It cannot see or touch the robot.** Every physical fact had to be measured by a person.
- **Confident drafts that were physically wrong.** It removed a "redundant" move that was actually how the arm reached the far rank.

TAKEAWAYS

1. **Measure, do not assume.** Every physical fact, from the arm's reach to each piece's grip height, had to be measured on the real robot.
2. **AI struggles with niche hardware.** It did not know the Dobot and made confident mistakes, so we gave it the documentation and checked everything it wrote.
3. **Design for recovery.** A missed grab or a bad camera read should not end the game: the camera double-checks and the screen asks a person to step in.